

# Case Study: E-Commerce — Spring Core + JdbcTemplate

## Objective

Build the repository layer for a simple e-commerce system using Spring's `JdbcTemplate`. DB connectivity (HikariCP, DataSource config) is already set up — focus is purely on writing repository methods.

## Entities

**Category:** `id`, `name`, `description`

**Vendor:** `id`, `name`, `email`

**Product:** `id`, `name`, `price`, `stockQuantity`, `category_id` (FK → Category), `vendor_id` (FK → Vendor)

## Tasks

### Task 1 — Insert Product with FK references

Implement `save(Product product)` in `ProductRepository`, inserting `category_id` and `vendor_id` correctly.

### Task 2 — Find Product by ID with joined data

Implement `findById(Long id)` returning a `Product` (or DTO) that includes the category name and vendor name — using a `JOIN` query and a `RowMapper`.

### Task 3 — Update stock quantity

Implement `updateStock(Long productId, int newQuantity)` as a targeted `UPDATE` (not

### Task 4 — Aggregate query

Implement `countProductsByVendor()` returning a `Map<String, Integer>` (vendor name → product count) using `JdbcTemplate.query` with a `GROUP BY + JOIN`, and a `RowMapper` to build the map.